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SUMMARY

Applied Scientist building LLM agents, with hands-on work in post-training, agent evaluation and synthetic data generation, and research in trustworthy ML and multi-agent systems. Published at ICLR 2026 and EMNLP 2025, with two years of production ML at NCR Voyix.

EXPERIENCE

SAP Labs

May 2026 – Present

Applied Scientist, Agentic AI

Palo Alto, CA

- Own model research for the SAP Joule NL-to-OData agent across synthetic data, SFT/RL, distillation, evaluation and serving.
- Built Terminator, a synthetic data engine that uses LLMs to generate schema-grounded agentic trajectories from real OData \$metadata, aimed at the query patterns the student model fails on, with a multi-stage pipeline plus online RL.
- Built the agent evaluation harness on ~1,100 tiered queries across S/4HANA, Ariba and SuccessFactors, pairing execution-based row and count matching with LLM judges and sorting every checkpoint into diagnosable failure buckets.
- A pass@k analysis based on RLVR incentivization found an RL capability ceiling and moved the roadmap to SFT on reasoning traces.
- Designed an execution-grounded RL reward with the live OData service as terminal reward, and built a failure-to-correction corpus that trains self-correction into the model weights.
- Distilled a Gemma 31B teacher into Gemma 4 E4B using LoRA, quantization and offline DAPO on teacher rollouts, reaching near-teacher accuracy with 7x faster inference.

Responsible Data Science (ReDS) Lab, Virginia Tech

Oct 2024 – May 2026

Graduate Research Assistant, advised by Dr. Ruoxi Jia

Blacksburg, VA

- Diagnosed multi-agent failures on GAIA and SWE-bench by backtracking OpenTelemetry execution traces to localize and categorize the causal failure span in long agentic trajectories.
- Amazon Nova Trusted AI Challenge. Built the LoRA/QLoRA and GRPO fine-tuning pipeline trading off code utility against security, plus a containerized 13-metric evaluation pipeline where LLM judges run alongside CodeQL, CodeGuru and Semgrep.
- Wired LLM-driven jailbreak attackers (PAIR, AutoDAN-Turbo, Crescendo-MT) into a red-teaming CI/CD pipeline, so each checkpoint was attacked again between training runs.
- Adversarial Deja Vu (ICLR 2026, with Princeton and Amazon AGI) compressed 16,901 adversarial skills from 32 jailbreak papers into 397 primitives, and training on their compositions cut StrongREJECT ASR on unseen post-cutoff attacks to 0.04–0.11.
- Confusion-Inducing Attack (EMNLP Main 2025, with Cisco Research) probes training-data memorization by steering LLMs into spikes of token-level entropy, reaching 22.2% verbatim extraction on Llama-2 70B.

NCR Voyix

Jul 2022 – Aug 2024

Software Engineer – Machine Learning

Hyderabad, India

- Built a LayoutLM document classifier over 450 OCR-noisy scanned documents in 9 classes, fine-tuning layout and token embeddings jointly to raise accuracy from 62% (rule-based extractor) to 96%.
- Wrote a Python log-analysis tool that clusters log spans and surfaces the first anomalous one, cutting regression localization in globally distributed CI/CD from ~50 min to ~4 min.
- Ran MLOps on Azure AKS with Spark, Kafka and Redis data pipelines, reducing pipeline latency by about 40%.
- Automated retraining and model versioning with MLflow and Airflow CI/CD, cutting retrain-to-deploy time from days to hours.
- Built Python REST services with monitoring and alerting for the production ML stack, and handled incident response for those services.

SELECTED PUBLICATIONS

- Adversarial Deja Vu: Jailbreak Dictionary Learning for Stronger Generalization. *ICLR 2026*.
- Retracing the Past: LLMs Emit Training Data When They Get Lost. *EMNLP Main 2025*.
- Data is All You Need (Almost): Iterative Synthetic Instruction Tuning for Secure Code Generation. *Amazon Nova AI Challenge 2025*.
- IDDAW: A Benchmark for Safe and Robust Segmentation of Drive Scenes in Unstructured Traffic and Adverse Weather. *WACV 2024*.

SKILLS

Agentic Systems Tool Calling, Multi-Agent Systems, LangGraph, MCP, RAG, Context Engineering, Langfuse, vLLM, SGLang

Post-Training SFT, RLHF, GRPO, DAPO, GiGPO, LoRA/QLoRA, Distillation, Quantization, RL Reward Design, Synthetic Data Generation

Agent Evaluation Eval Harnesses, LLM-as-a-Judge, pass@k Analysis, Benchmark Design, Failure Analysis, OpenTelemetry, Red Teaming

Machine Learning PyTorch, Hugging Face, Transformers, NLP, Reinforcement Learning, Deep Learning, Experimental Design, A/B Testing

Languages Python, C++, SQL, Bash

Infrastructure Docker, Kubernetes, Azure (AKS), AWS (EC2, S3, Bedrock), Kafka, Spark, MLflow, Airflow, CI/CD, Git

PROJECTS

Agentic RAG Code Assistant with MCP Integration

Python, LangGraph, FAISS, vLLM, Filesystem/GitHub MCP

- Built a LangGraph agent that answers natural-language questions over local codebases, GitHub repositories and commit history, calling Filesystem and GitHub MCP tools and a FAISS store of chunk-level embeddings.
- Improved retrieval context by moving from fixed-size to AST-aware chunking and re-selecting the embedding model, measured with recall@k and MRR on a held-out question set.

AWARDS AND LEADERSHIP

- **Amazon Nova AI Challenge 2025**, Team HokieTokie, code generation track.
- **Technical Lead, ML4E Club**, Virginia Tech. Led ML initiatives and mentored students on research projects.

EDUCATION

Virginia Tech

Aug 2024 – May 2026

M.S. in Computer Engineering - GPA 3.67/4.0

Blacksburg, VA

National Institute of Technology, Rourkela

Jul 2018 – Jun 2022

B.Tech in Electrical Engineering

Rourkela, India